

Mitchell Tyler Scott

(Last updated: May 27, 2026)

Curriculum Vitae

CONTACT INFORMATION
Department of Mathematics
Emory University
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EDUCATION
Emory University, Atlanta, Georgia, USA
Ph.D., Department of Mathematics, expected in May 2027
M.S., Department of Computer Science, expected in August 2026
Advisor: Professor Yuanzhe Xi
Tenative thesis title: Data-driven Preconditioners for Data Science Problems
Tufts University, Medford, Massachusetts, USA
M.S., Department of Mathematics, May 2023
Advisors: Professor Misha Kilmer, Professor Xiaozhe Hu
Thesis: A Tale of Two Tensors: Using Hierarchical and Block Low Rank Matrices to Make Preconditioners and Save Storage
Cornell University, Ithaca, New York, USA
B.S., Department of Biological Engineering, December 2020
Minors: Pure Mathematics, Applied Mathematics
Thesis: Designing and Optimizing a Protocol for Whole-Ovary Vitrification

RESEARCH INTERESTS
Preconditioners for Stochastic Optimization
Numerical Linear Algebra
High Performance Scientific Computing

PUBLICATIONS
Journal Articles:
1. [M.T. Scott](#), T. Xu, Z. Tang, A. Pichette-Emmons, Q. Ye, Y. Saad, and Y. Xi, *Designing Preconditioners for SGD: Local Conditioning, Noise Floors, and Basin Stability* Transactions of Machion Learning Research, June 2026 [[arXiv](#)]
Theses:
2. [M.T. Scott](#) “A Tale of Two Tensors: Using Hierarchical and Block Low Rank to Make Preconditioners and Save Storage”, Tufts University, Medford, MA, 2023 [[pdf](#)]
1. [M.T. Scott](#) “Designing and Optimizing a Protocol for Whole-Ovary Vitrification”, Cornell University, Ithaca, NY, 2020 [[pdf](#)]
Peer Reviewed Teaching Activities:
1. [M.T. Scott](#) (2025) *Markov Chains in Public Transportation*, Submitted October 7, 2025, Accepted October 26,2025, from <https://serc.carleton.edu/309394>

ON-GOING RESEARCH
Structured matrices, arising from the abstraction of real world physical systems modelled by discretized partial differential equations, are prevalent in fluid dynamics, computational finance, and image processing. Current ways to store and solve problems using these matrices can be slow. My current research is interested in expointing hidden structure so that we can construct robust and novel approximations of the Hessian to be used in nonlinear preconditioning for scientific and machine learning problems. More technical research topics include: iterative methods, multilinear algebra, tensor-based decomposition, fractional and nonlinear PDEs, and preconditioning.

CONFERENCES, WORKSHOPS AND TALKS
Contributed Talks:

7. *What Makes a Good Data Science Preconditioner?*,
Preconditioning techniques for scientific and industrial applications, Edinburgh, Scotland, UK, May 27, 2026.
6. *Block Subset Selection based on Randomized QR with Column Pivoting: A $\star\mathbf{M}$ Framework*,
27th Conference of the International Linear Algebra Society, Blacksburg, VA, May 22, 2026.
5. *What Makes a Good Data Science Preconditioner?*,
Southeast Applied and Computational Math Student Workshop, Atlanta, GA, April 3, 2026.
4. *Preconditioning Stochastic Gradient Descent: Theoretical Guarantees & Novel Framework*,
Southeast Applied and Computational Math Student Workshop, Auburn, AL, April 5, 2025.
3. *Preconditioning Stochastic Gradient Descent: Theoretical Guarantees & Novel Framework*,
Atlanta SIAM Student Conference, Atlanta, GA, March 28, 2025.
2. *Constructing Hierarchical Matrices through Recursive Tensor Decomposition*,
Conference on Fast Direct Solvers, West Lafayette, IN, November 4, 2023.
1. *Discovering Hierarchical Matrix Structure Through Recursive Tensor Decomposition: Joint Mathematics Meeting*, Boston, MA, January 4, 2023.

Contributed Posters:

7. *What Makes a Good Preconditioner for Data Science?*. Georgia Scientific Computing Symposium, GSCS 26, Athens, GA, February 21, 2026
6. *What Makes a Good Preconditioner for Data Science?*. Randomized Numerical Linear Algebra Workshop, ICERM, Providence, RI, February 5, 2026.
5. *Accelerating and Stabilizing SGD via Preconditioning: A Unified Analysis*. Learning about Laney Showcase, Emory University, Laney Graduate School, Atlanta, GA, Sept 29, 2025.
4. *Accelerating Multigrid Prolongation via Bipartite Graph Attention Networks*. Summer Student Showcase, Lawrence Livermore National Lab, Livermore, CA, August 6, 2024
3. *Fisher-informed Nonlinear Truncated Generalized Conjugate Residual Optimizer*. Georgia Scientific Computing Symposium, GSCS 25, Atlanta, GA, February 8, 2025
2. *Acceleration Methods for Scientific and Data Science Applications*. SIAM Mathematics of Data Science, SIAM MDS 24, Atlanta, GA, October 25, 2024
1. *Quantized Tensor Trains for Solving Maxwell's Equations with Spectral Methods*. Computing Sciences Poster Session, Berkeley Lab, Berkeley, CA, August 6, 2024

Student Seminar Talks:

7. *All Aboard the Tensor Train: Using Spectral Methods in the Tensor Train Framework to solve Maxwell's Equation*, Emory University Discussions in Scientific Computing, Atlanta, GA, February 24, 2025.
6. *All Aboard the Tensor Train: Using Spectral Methods in the Tensor Train Framework to solve Maxwell's Equation*, Tufts University SIAM Seminar, Medford, MA, February 19, 2025.
5. *Integration over Lie Groups, with applications to random matrix theory*, Emory University Research in Algebra and Number Theory, Atlanta, GA, April 11, 2024.
4. *A Tale of Two Tensors: Using Hierarchical and Block Low Rank to Make Preconditioners and Save Storage*, Emory University Discussions in Scientific Computing, Atlanta, GA, October 6, 2023.
3. *It's Tensor Time!: A Computational Framework for Analyzing Structured Matrices*, Tufts University Monday Math Meeting, Medford, MA, September 12, 2022.
2. *Representation Schemas for Visualizing Quantum Algorithms*, Tufts University Quantum Computing Reading Group, Medford, MA, April 11, 2022.
1. *Special Families of Matrices used in Quantum Algorithms*, Tufts University Quantum Com-

puting Reading Group, Medford, MA, February 28, 2022.

Workshops and Conferences Attended

18. [Preconditioning techniques for scientific and industrial applications](#), Precond 26, ICMS, Bayes Centre, Edinburgh Scotland, UK, May 27-29, 2026.
17. [27th Conference of the International Linear Algebra Society ILAS 2026](#), Virginia Tech, Blacksburg, VA, May 18-22, 2026.
16. [Southeast Applied and Computational Math Student Workshop '26](#), Emory University, Atlanta, GA, April 3-4, 2026.
15. [Georgia Scientific Computing Symposium '26](#), University of Georgia, Athens, GA, February 21, 2026.
14. [Randomized Numerical Linear Algebra](#), ICERM, Brown University, Providence, RI, Feb. 2-6, 2026.
13. [Teaching Computation with MATLAB \(and GenAI\)](#), SERC, Carleton College, Northfield, MN, October 26-28, 2025.
12. [Randomized Numerical Linear Algebra Research Collaboration Workshop](#), IPAM, Los Angeles, CA, August 10-16, 2025.
11. [Southeast Applied and Computational Math Student Workshop '25](#), Auburn University, Auburn, AL, April 5-6, 2025.
10. [Georgia Scientific Computing Symposium '25](#), Georgia Institute of Technology, Atlanta, GA, February 8, 2025.
9. [SIAM Mathematics of Data Science](#), MDS 25, Atlanta, GA, October 21-25, 2024.
8. [Georgia Scientific Computing Symposium '24](#), Emory University, Atlanta, GA, February 24, 2024.
7. [Conference on Fast Direct Solvers](#), Purdue University, West Lafayette, IN, November 4-5, 2023.
6. [Acceleration and Extrapolation Methods](#), ICERM, Providence, RI, July 24-28, 2023.
5. [Qiskit Global Summer School 2023: Theory to Implementation](#), IBM, July 17-28, 2023.
4. [Joint Mathematics Meetings](#), Boston, MA, January 4-7, 2023.
3. [Geometry and Analysis Seminar for Boston Area Graduate Students](#), Massachusetts Institute of Technology, Cambridge, MA, October 29-30, 2022.
2. [Qiskit Global Summer School 2022: Quantum Simulations](#), IBM, July 18-29, 2022.
1. [Geometry and Analysis Seminar for Boston Area Graduate Students](#), Massachusetts Institute of Technology, Cambridge, MA, November 6-7, 2021.

RESEARCH EXPERIENCE

Lawrence Livermore National Laboratory - Livermore, CA

Visiting Graduate Student Researcher: May 2025 - September 2025

Lawrence Berkeley National Laboratory - Berkeley, CA

Visiting Graduate Student Researcher: May 2024 - August 2024

TEACHING EXPERIENCE

Emory University

Instructor of Record

- MATH 190 - Mathematics of Sustainability (Spring 2026)
- MATH 111 - Calculus I (Fall 2025)

Teaching Assistant

- MATH 210 - Advanced Calculus for Data Science (Fall 2026): **Head TA**
- MATH 210 - Advanced Calculus for Data Science (Spring 2025): **Head TA**
- MATH 210 - Advanced Calculus for Data Science (Fall 2024)
- MATH 315 - Numerical Analysis (Spring 2024)

Directed Reading Program Mentor

- Introduction to Stochastic Jump Processes (Spring 2026)
- Introduction to Stochastic Processes (Spring 2026)
- Advanced Game Theory: Algorithms and Voting Systems (Fall 2025)
- Introduction to Stochastic Calculus (Spring 2025)
- Introduction to Functional Analysis and Finite Elements (Fall 2024)
- Introduction to Stochastic Processes (Spring 2024)
- Introduction to Iterative Methods for Inverse Problems (Fall 2023)
- Introduction to Stochastic Processes (Fall 2023)

Course Assistant

- MATH 351 - Partial Differential Equations (Fall 2023)

Tufts University

Teaching Assistant

- MATH 126 - Numerical Linear Algebra (Spring 2023)

Directed Reading Program Mentor

- Introduction to Mathematical Control Theory (Fall 2022)

Course Assistant

- MATH 125 - Numerical Analysis (Fall 2022)
- MATH 32 - Calculus I (Spring 2022)

Cornell University

Undergraduate Teaching Assistant

- BEE 2600 - Introduction to Biological Engineering (Fall 2018)

HONORS AND
AWARDS

Conference Travel Grant,

Preconditioning techniques for scientific and industrial applications, International Centre for Mathematical Sciences, £900 for Summer 2026

Conference Travel Grant,

International Linear Algebra Society, Virginia Tech, \$300 for Spring 2026

Conference Travel Grant,

Institute of Computational and Experimental Research in Mathematics, Randomized Numerical Linear Algebra, \$850 for Spring 2026

Conference Travel Grant,

Carleton College, Science Education Resource Center at Carleton College, \$500 for Fall 2025

Summer Student Showcase Best Presenter,

Lawrence Livermore National Laboratory, Data Science Institute, 2025.

Summer Research Presentation Superlative,

Sustainable Horizons Institute, 2024.

Piedmont Teaching Fellowship in Sustainability and Curriculum Development,

Emory University, \$1,000 for AY 2024 - 2025, 2025 - 2026.

Conference Travel Grant,

Institute of Computational and Experimental Research in Mathematics, Acceleration and Extrapolation Methods, \$850 for Summer 2023

Research Conference Presenter Grant,
Tufts University, Graduate School of Arts and Science, \$600 for AY 2022 - 2023

Department of Mathematics Scholarship,
Tufts University, \$20,000 for AY 2022 - 2023

The Fuertes Medal Memorial Prize for Public Speaking,
Cornell University, College of Engineering, \$3,000 for AY 2019 - 2020.

MEMBERSHIPS	American Mathematical Society (AMS) Society for Industrial and Applied Mathematics (SIAM) Spectra!
SERVICE	Local Organizing Committee: Southeast Applied & Computational Math , April 3-4, 2026, Emory University, Atlanta, GA.
DEPARTMENTAL SERVICE	Emory University Officer: Directed Reading Program Steering Committee, 2023 - Present President: SIAM Chapter, 2025 - 2026 Risk Manager: SIAM Chapter, 2025 - 2026 Site Coordinator: American Mathematics Competition (AMC-8, American Invitational Mathematics Examination), 2025 - Present
STEM OUTREACH LANGUAGE SKILLS	Mary STEAM English: native speaker Programming: MATLAB, JULIA, PYTHON, C/C++ Computer: L ^A T _E X, Microsoft Office, HTML, CSS